

I am a consultant and machine learning researcher with a dual background in Internal Medicine and Computer Science, currently appointed as a postdoctoral researcher at Vita-Salute San Raffaele University, Milan, and visiting researcher at the University of Cambridge. My work bridges medicine and computational methods, focusing on reproducible pipelines for the acquisition, preprocessing and use of routinely collected data from electronic medical records for risk prediction modelling and clinical decision support according to expected health trajectories. I hold experience in teaching at undergraduate/graduate level, authoring and editing peer-reviewed articles, and am used to engage with various stakeholders, including clinicians, researchers and industry partners.

EDUCATION

- 2023-2025 **PhD in Health Data Science**, cum Laude, Vita-Salute San Raffaele University/IRCCS San Raffaele Hospital, Milan, Italy (Supervisor: Prof Patrizia Rovere Querini – Milan, Co-Supervisor: Prof Cecilia Mascolo – Cambridge)
- Led real-world multimodal data collection, integration between silos, quality control, and analysis of complex multimorbid internal medicine patients (MED-Cli, NCT05780099, ~2000 patients), *paper under review*
 - Implemented a clinical trial to acquire and preprocess data from Electronic Medical Records of Type 2 diabetes mellitus patients; performed quality control and applied Machine Learning for predictive tasks (AI-TRYDIA, NCT06280729, ~9000 patients) (*Front Dig Health paper*)
 - Visiting Researcher at the Department of Medicine/Cambridge University Hospital under the supervision of Dr Ari Ercole, Chief Informatics Officer:
 - Developed ML pipelines to leverage real-world data of 17000 patients at Cambridge University Hospital for external validation experiments
 - Attended the 2nd and 4th summer school in AI and Machine Learning for Medicine by the Cambridge Centre for AI in Medicine and the 1st summer school in AI for Healthcare by the Italian Association for Artificial Intelligence
 - Covered topics such as ML and deep learning fundamentals, model evaluation and interpretability, Large Language Models and their use for research and coding, and causal inference
 - Co-designed with Microsoft and partners the S-RACE cloud-based multilayer platform for an end-to-end clinical data science pipeline in AzureML Studio (*npj Digital Medicine paper*)
 - Collected and curated three different clinical datasets for research purposes (*papers: COVID19 voice-based detection, lung cancer, COVID19 prognostication*)
- 2018-2023 **Internal Medicine Specialty**, cum Laude, Vita-Salute San Raffaele University, Milan, Italy
- Thesis project: instituted the Primary Care-Hospital Embedding Prospective, Multicentric, Observational Clinical Trial to introduce integrated care at IRCCS San Raffaele Hospital (PRIME, NCT05723185)
- 2011-2017 **Degree in Medicine**, cum Laude, International MD Program, Vita-Salute San Raffaele University, Milan, Italy
- Student intern in the Unit of Haematology and Bone Marrow Transplant
 - Wet lab thesis project: performed fluorescence cytometry screening in Jurkat T cell models of optimal Chimeric Antigen Receptor constructs. Supervisor: Prof Fabio Ciceri – Co-Supervisor: Dr Attilio Bondanza

WORK EXPERIENCE

- 2025-Present **Postdoctoral Research Associate, Artificial Intelligence team, Vita-Salute San Raffaele University, Milan, Italy**
- Leading an interdisciplinary team to study routinely collected data preprocessing pipelines and machine learning modelling for the prediction of type 2 diabetes mellitus control and of hospitalisation outcomes (NRRP funded D3-4H project, PI: Prof Carlo Tacchetti)
- 2023-Present **Internal Medicine Consultant, Emergency Department, IRCCS San Raffaele Hospital, Milan, Italy**
- Caring for ~900 patients/year accessing the lower intensity branch of the ED

AWARDS AND ACHIEVEMENTS

- Awarded a **merit-based scholarship** for five of six years of the medical school, covering approximately €18000 per annum in tuition fees, plus extras for accommodation and living expenses
- Secured €9790 from 181 contributors through an **independent crowdfunding initiative** (2015), to support tuition and maintenance costs during the year not covered by institutional funding
- **Best Oral Communication** (1/10), 'AI-SCoRE (Artificial Intelligence-SARS CoV2 Risk Evaluation): a fast, objective and fully automated platform to predict the outcome in COVID-19 patients', 123rd National Congress of the Italian Society of Internal Medicine, Oct 2022, Rome
- **Best Oral Communication** (1/6), 'HomeDoc: helping physicians during the COVID19 crisis through an eHealth platform and machine learning', 121st National Congress of the Italian Society of Internal Medicine, Oct 2020, Held Virtually

PUBLICATIONS, CONFERENCES AND WORKSHOPS

- 5 first-author, 12 co-authored publications
- 6 refereed conference presentations with peer-reviewed abstract submission at international and national conferences

Selected publications:

- **M. Montagna**, A. S. Rabadzhiev, et al. 'From raw data to actionable insights: preprocessing real-world data for machine learning in diabetes care', *Frontiers in Digital Health*, 2026

- A. Traverso, ..., **M. Montagna** et al. 'Powering responsible artificial intelligence with high-quality real-world data: The S-RACE platform for scalable, multi-specialty clinical research', npj Digital Medicine, 2026
- C. Mascolo, T. Xia, E. Carlsson, M. Kågebäck, **M. Montagna**, John Ji. 'Uncovering Respiratory Effects of Short-term PM2.5 Exposure through Global Nocturnal Cough Monitoring', Nature Health, under review
- J. Han, **M. Montagna**, ..., C. Mascolo. 'Evaluating Listening Performance for COVID-19 Detection by Clinicians and Machine Learning: Comparative Study', Journal of Medical Internet Research, 2023
- F.R. Ogliari, A. Traverso, S. Barbieri, **M. Montagna**, et al. 'Exploring machine learning tools in a retrospective case-study of patients with metastatic non-small cell lung cancer treated with first-line immunotherapy: A feasibility single-centre experience', Lung Cancer, 2025
- **M. Montagna**, F. Chiabrando, R. De Lorenzo, P. Rovere Querini. 'Impact of Clinical Decision Support Systems on Medical Students' Case-Solving Performance: Comparison Study with a Focus Group', JMIR Medical Education, 2025

Selected conferences and workshops:

- **Invited Speaker**, 'Real World Data-driven medicine: opportunities and challenges to improve everyday clinical practice', 3rd Research and Innovation in Internal Medicine, Jun 2025, Bucharest
- **Invited Speaker**, Audio based AI for Respiratory Health Monitoring, Mar 2025, University of Cambridge
- 'A diagnosis-independent frailty index at admission improves risk stratification in hospitalized internal medicine patients', 24th European Congress of Internal Medicine, Mar 2026, Vienna
- Lecturer and organizer - Pre-Conference Workshop 'AI tools for clinical research: the basics for internists', 23rd European Congress of Internal Medicine, Mar 2025, Florence ~30 attendees
- 'Leveraging Real World Data with Machine Learning to predict long term HbA1c changes in Type 2 Diabetes Mellitus', 23rd European Congress of Internal Medicine, Mar 2025, Florence
- Delivered a **joint webinar** of the Young Internists of EFIM and SIMI, 'The (unexpected) fundamentals of Artificial Intelligence for Internal Medicine', Feb 2025, ~60 attendees, >340 views of the recording on YouTube

SKILLS AND INTERESTS

Research: case report form and clinical database management; ethics&data governance; pipelines for routinely collected clinical data acquisition, integration, quality control and preprocessing; exploratory data analysis; machine learning model training-validation and external testing; protocol/clinical trial/grant writing; interdisciplinary teamworking; oral presentation; seminar organization/delivery

IT: Python, MS Visual Studio Code, MS Office, Adobe suite, Canvas and other conventional software

Languages: Italian, as mother tongue; English, fluent in written and spoken language

Working to Deadlines & Time Management: used to balance concurrent research, teaching and clinical activities (es overlapping PhD Year 1 and Residency Year 5 required completion of research assignments while attending lectures and completing clinical rotations)

Interests: competitive track and field athlete, sourdough baker, drummer, photographer and video maker

TEACHING EXPERIENCE & PUBLIC OUTREACH

- 2025-present Tutor of internal medicine and physical examination clerkships at the simulation centre, ~150 students
- 2024-present Lecturer and examiner of the Residency Program of Internal Medicine | **Mentor** for three residents each year
- 2024&2025 Volunteer at Milan's MEETmeTonight Science Festival and BergamoScienza Festival
- Nov 2024 Delivered an invited **public seminar** on Artificial Intelligence in Healthcare engaging a non-specialist audience to promote understanding of AI applications and ethical implications in medicine, Castelbuono (PA), Italy
- 2023-present **Thesis Co-Supervisor** of 2 resident physicians, 4 MD students, 2023-present
- 2023-present Lecturer of Internal Medicine, course and material organizer and marking, ~120 students

VOLUNTARY EXPERIENCE

- 2017-Present **Scout Leader (Wood Badge certified) – AGESCI**
Leading and mentoring youth groups | Designing and supervising experiential learning activities

EVIDENCE OF ESTEEM

- Member: Telemedicine, Innovative Technologies and Digital Health Working Group – European Federation of Internal Medicine, from 2023
 - Co-author of the position paper: I. Said-Criado et al. 'Advancing Toward P6 Medicine: Recommendations for Integrating Artificial Intelligence in Internal Medicine', Clinics and Practice, 2025
- Member: Artificial Intelligence and Digital Therapeutics Working Group – Italian Society of Internal Medicine, from 2024
 - Co-first-author of the position paper: C. Balsano et al. 'Artificial Intelligence in Medicine: a position paper by the Italian Society of Internal Medicine', Internal and Emergency Medicine, 2025
- **Founder:** Italian Society for Artificial Intelligence in Medicine (SIAM), from 2022
- **Associate Editor:** JMIR Medical Education, from 2025
- Active reviewer for journals: Internal and Emergency Medicine, Journal of endocrinological investigation from 2023
- **Representative:** Young Internist of Lombardy in the National Society of Internal Medicine (SIMI), from 2024
 - Co-Organizing the annual conference of the Lombardy Branch, Italian Society of Internal Medicine, ~150 attendees
 - Co-Organizing the annual Science Communication course, Italian Society of Internal Medicine, ~20 attendees